COURSE PROJECT: MACHINE LEARNING

Semester I – Academic Year 2025–2026

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Version History

Version	Data	Update Description
1.0	2025-31-08	Initial release of the document
1.1	2025-09-01	Updated academic year and semester

1 Introduction

The course project in Machine Learning is designed to help students apply the theoretical knowledge they have learned to solve real-world problems across various types of data. Through building data processing pipelines, performing feature extraction, training models, and evaluating results, students will gain a solid understanding of the fundamental steps involved in developing a complete machine learning system.

The project covers three common types of data: tabular data, text data, and image data. Each data type has its own characteristics, requiring students to select appropriate processing techniques and models. As a result, students not only reinforce their theoretical knowledge but also develop practical skills, analytical thinking, and the ability to write scientific reports.

2 Overview of the Machine Learning Pipeline

In machine learning, a system is typically developed through a sequence of steps known as a *pipeline*. The pipeline describes the entire process, from raw data to building predictive models and evaluating their performance. Currently, there are two common approaches: the traditional machine learning pipeline and the deep learning pipeline.

The traditional machine learning pipeline typically includes:

1. **EDA (Exploratory Data Analysis):** performing descriptive statistics, data visualization, detecting missing values, outliers, and distribution trends.
2. **Data preprocessing:** cleaning data, handling missing values, normalization or standardization, encoding categorical variables, and splitting data into training and testing sets.
3. **Feature extraction and selection:** applying different techniques depending on the data type, for example:
    - **Text data:**

      – Traditional: Bag-of-Words, TF-IDF, n-gram

      – Deep learning: RNN/LSTM, Transformer (BERT, GPT)

    - **Image data:**

      – Traditional: HOG (Histogram of Oriented Gradients), SIFT, PCA for dimensionality reduction after vectorization

      – Deep learning: CNN (Convolutional Neural Networks), ResNet, VGG, EfficientNet, Transformer-based models (ViT, Swin Transformer)

    - **Tabular data:**

      – Traditional: statistical features (mean, std), normalization (z-score, min-max), categorical encoding (one-hot encoding, label encoding)

      – Deep learning: MLP (Multilayer Perceptron), TabNet, NODE (Neural Oblivious Decision Ensembles)

4. **Model training:** applying traditional machine learning algorithms such as Logistic Regression, SVM, k-NN, Random Forest, and Naive Bayes.
5. **Model evaluation:** using metrics such as accuracy, precision, recall, and F1-score.

The modern deep learning pipeline is often simpler in terms of feature extraction due to the representation learning capability of neural networks:

1. **EDA (Exploratory Data Analysis):** data visualization, descriptive statistics, distribution checking, and detecting missing or abnormal data.

2. **Basic data preprocessing:** normalization, image resizing, text tokenization, and sequence padding.
3. **Neural networks for feature learning:** CNNs for images, RNN/LSTM for sequential data, and Transformers for text and images (ViT, BERT).
4. **End-to-end training:** the entire model is optimized simultaneously using gradient descent.
5. **Evaluation and deployment:** using appropriate metrics (accuracy, F1-score, BLEU, IoU) and deploying the model into real-world applications.

Within the scope of this course, students are required to implement the pipeline using the traditional approach. Groups that additionally implement a deep learning pipeline and conduct a comparison will receive bonus points to encourage exploration and creativity.

3 Objectives

The course project is designed with the following specific objectives:

- To understand and apply the traditional machine learning pipeline, including data preprocessing, feature extraction, model training, and evaluation.
- To develop practical skills in implementing machine learning models on different types of data: tabular, text, and image.
- To enhance the ability to analyze, compare, and evaluate the performance of machine learning models using appropriate metrics.
- To improve programming, experimentation, and scientific report writing skills.
- **Required:** implement a traditional machine learning pipeline on at least one type of data.
- **Encouraged:** if possible, implement an additional deep learning pipeline to compare with the traditional approach. Groups that include this extension will receive bonus points.

4 Content

The course project consists of three topics corresponding to three common types of data: tabular data, text data, and image data. Each group or individual is required to choose at least one topic to implement. For each selected data type, students must complete all steps of the traditional machine learning pipeline, including EDA, data preprocessing, feature extraction, model training, and evaluation.

Conducting EDA (Exploratory Data Analysis) is a mandatory requirement, as it helps students thoroughly understand the characteristics of the data before performing analysis and model development. In addition, if students implement an extra deep learning pipeline to compare with the traditional pipeline, their report will receive bonus points. This encourages students to broaden their knowledge and gain a more comprehensive view of both approaches in machine learning.

4.1 Assignment 1: Machine Learning with Tabular Data

Students will be divided into groups (see details in the *Working Format* section), and each group must select an appropriate dataset to implement the assignment, subject to the following constraints:

- Groups are not allowed to choose the same dataset.
- The dataset must contain missing values so that students can practice imputation techniques.
- The dataset must include categorical variables so that students can apply encoding techniques.
- The sample size must be sufficiently large for the pipeline to be meaningful; the specific choice should be discussed directly with the instructor in class.

Each group is required to build a traditional machine learning pipeline for tabular data. The pipeline should be designed to allow configuration of techniques and parameters at each step. For example:

- **Scaling:** choose between MinMaxScaler or StandardScaler; if MinMaxScaler is selected, students need to configure the *feature_range*.
- **Dimensionality reduction:** choose PCA with different retained variance levels (e.g., 90%, 95%, etc.).
- **Models:** choose among Logistic Regression, SVM, Random Forest, and compare their performance.

The final deliverables must include: an EDA analysis report, a description of the pipeline, the experimented configurations and parameters, and a comparison of performance across different settings.

4.2 Assignment 2: Machine Learning with Text Data

Students will be divided into groups (see details in the *Working Format* section), and each group must select a text dataset to implement the assignment, subject to the following constraints:

- Groups are not allowed to choose the same dataset.
- The dataset must have diverse content (e.g., multiple topics or multiple classification labels).
- The sample size must be sufficiently large for the model to be meaningful; the specific choice should be discussed directly with the instructor in class.

In this assignment, students are required to implement at least two feature extraction methods for text data:

- **Traditional methods:** Bag-of-Words (BoW), TF-IDF, or n-gram.
- **Modern methods (deep learning embeddings):** using deep learning models to represent text. Suggested approaches include:

    - Word2Vec or GloVe (pre-trained embeddings).
    - BERT, RoBERTa, or DistilBERT (Transformer-based contextual embeddings).

Students need to generate embeddings and save the results in .npy or .h5 format, then use these embeddings as input for traditional classification models or simple neural networks.

The pipeline should be designed flexibly, allowing configuration of preprocessing steps (e.g., stopword removal, tokenization, padding), selection of embedding techniques, and choice of classification models (Naive Bayes, Logistic Regression, SVM, or fine-tuning BERT).

The final deliverables must include: an EDA report (e.g., word count statistics, text length distribution, word frequency), a description of the pipeline, the experimented parameters, and a comparison between traditional and modern approaches.

4.3 Assignment 3: Machine Learning with Image Data

Students will be divided into groups (see details in the *Working Format* section), and each group must select an image dataset to implement the assignment, subject to the following constraints:

- Groups are not allowed to choose the same dataset.
- The dataset must contain a sufficiently large and diverse number of images for meaningful training; this should be discussed and approved by the instructor beforehand.
- The image size and quality should be appropriate for processing in the working environment (e.g., Google Colab or personal computers).

In this assignment, students are required to perform basic EDA for image data (e.g., image size, number of color channels, label distribution, number of samples per class), and then build a pipeline with the following steps:

- **Modern feature extraction (deep learning features):** use pretrained deep learning models to extract image features. Suggested models include:
    - **CNN-based models:** VGG16, ResNet, EfficientNet
    - **Transformer-based models:** Vision Transformer (ViT), Swin Transformer

These extracted features must be saved in .npy or .h5 format and used as input for subsequent classification models.

The pipeline should be flexible, allowing configuration of different steps (e.g., image resizing, choice of pretrained model for feature extraction, selection of classification models such as Logistic Regression, SVM, or Random Forest).

The final deliverables must include: an EDA report, a description of the pipeline, the saved features, and a comparison of performance across different deep learning model choices.

5 Deliverables

All submissions must include a front-end implemented as a Google Colab Notebook. Students may (and are encouraged to) develop additional supporting modules in Python, then import and use them directly within Colab.

The submitted Colab notebook must satisfy the following requirements:

- When selecting **Runtime → Run all**, the entire notebook must execute successfully without errors.
- It must not mount any personal cloud storage services (e.g., Google Drive, Dropbox, etc.).
- Data must be downloaded from public sources (with explicit links provided in the notebook) and extracted directly into the Colab environment.
- All data preparation steps and installation of additional libraries (if any) must be clearly specified and executed automatically within the notebook.

In addition to the notebook, the submission must include:

- **PDF report:** presenting EDA, pipeline design, experiments conducted, result comparisons, and analysis.
- **Extracted feature files** (e.g., embeddings, feature vectors) saved in .npy or .h5 format, either included in the submission or provided via download links.
- **Clear folder structure** in the submitted .zip file, including: notebooks/, modules/, reports/, and features/.
- Students are encouraged to upload all source code and materials to GitHub and share the repository link in both the Colab notebook and the report. This will receive bonus points as it demonstrates good practices in code management and scientific dissemination. The GitHub repository README must include:
    - Course name, course code, semester, and academic year
    - Instructor information
    - Team member details: full name, student ID, and email
    - Project objectives
    - Instructions for running the notebook (required libraries, how to download data)
    - Project folder structure
    - Links to the PDF report and Colab notebook (if available)

6 Working Format

Students are required to form groups of 1–5 members. Each group will complete the entire course project (including any extensions, if applicable) according to the registered group.

- **Group registration:** Groups must register their group name and member list in the Google Sheet shared on the course LMS.
- **Supporting evidence:** During the project, groups are encouraged to maintain evidence of collaboration (e.g., images, meeting recordings, discussion notes, etc.) to demonstrate actual teamwork.

- **Task allocation:** In the final report, each group must include a detailed task allocation table for all members, along with their percentage of contribution. Each member's score will be calculated as:

Individual Score = Group Score x Contribution Percentage (%)

7 Evaluation Criteria

The course project will be assessed based on the following criteria:

- **Completion of the traditional pipeline (40%):** Includes all required steps such as EDA, data preprocessing, feature extraction, model training, and evaluation.
- **Quality of analysis and experiments (25%):** Comparing multiple configurations, explaining results, and analyzing strengths and limitations.
- **Report quality (20%):** Clear presentation with tables, figures, task allocation, and individual contribution percentages.
- **Completeness of submission (10%):** The Colab notebook runs successfully using *Run all*, has a clear folder structure, and feature files are saved in the correct format.
- **Teamwork (5%):** Includes evidence of group collaboration (e.g., meeting photos/videos, discussion records).

Additional bonus points may be awarded:

- **Deep learning pipeline:** +5% for implementing and comparing with the traditional pipeline.
- **Publication on GitHub:** +5% for a well-organized repository with a README that includes course information, instructor details, team members, usage instructions, and links to the report/Colab notebook.

Note: Throughout the course, groups are required to submit periodic progress reports according to the schedule announced by the instructor. These reports serve as a basis for evaluating participation, progress, and task distribution within the group.

8 Submission Deadline & Method

- **Submission method:** The course project must be submitted via the Google Drive folder created and shared by the instructor with the entire class.
- **Deadline:** The specific submission deadline will be announced on the course LMS.

Extension for Honours Engineering Program

In addition to the main course project, groups from the Honors Engineering Program are required to complete an additional extension. Each group (keeping the same members as the main project) must choose **one of the following three topics** to implement:

1. Bayesian Network (BN)

- Implement a Bayesian Network model and develop inference algorithms, including exact inference and basic approximate inference.
- **Application:** solve a representative problem, such as classification or prediction based on causal relationships in data.
- The instructor will provide a Python library for working with DAGs and performing topological sorting, allowing students to focus on implementing inference algorithms.

2. Hidden Markov Model (HMM)

- Implement core algorithms: Forward, Viterbi, and (optionally) Baum–Welch for parameter training.
- **Application:** solve a typical sequential problem, such as character recognition, sequence labeling, or signal analysis.

3. Conditional Random Field (CRF)

- Implement a CRF model and basic inference algorithms.
- **Application:** solve a sequence labeling problem, such as Named Entity Recognition (NER) in text or sequence segmentation.

For the selected topic, each group must:

- Build the model by implementing it or using appropriate libraries.
- Evaluate the model through experiments and analyze its strengths and limitations.
- Present a representative application in the report and illustrate it with experimental results.